

Experiment 21: Swap the Cases of a Character Column in a DataFrame

Aim

To write a Pandas program to swap the uppercase and lowercase characters of a specified column in a given DataFrame.

Procedure

- Import the Pandas library.
- Create a DataFrame containing a character/string column.
- Select the required character column.
- Use the `swapcase()` function to interchange uppercase and lowercase letters.
- Display the original and modified DataFrame.

CODE

```
import pandas as pd

# Create DataFrame
data = {
    'Name': ['Alice', 'BOB', 'ChArLiE', 'David']
}

df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

# Swap the cases of the Name column
df['Name'] = df['Name'].str.swapcase()

print("\nDataFrame after swapping cases:")
print(df)
```

OUTPUT

```
Original DataFrame:
  Name
0  Alice
1   BOB
2 ChArLiE
3  David

DataFrame after swapping cases:
  Name
0  aLiCE
1   bob
2 cHaRlIe
3  dAVID
```

RESULT

The uppercase characters were converted to lowercase and the lowercase characters were converted to uppercase successfully using Pandas `str.swapcase()`.

Experiment 22: Draw a Line with Labels and Title

Aim

To write a Python program using Matplotlib to draw a line graph with suitable labels for the x-axis, y-axis, and a title.

Procedure

- Import the `matplotlib.pyplot` library.
- Define the x-axis and y-axis values.
- Plot the values using the `plot()` function.
- Add suitable labels to the x-axis and y-axis.
- Add a suitable title to the graph.
- Display the graph using `show()`.

CODE

```
22.py - C:/Users/subha/OneDrive/Desktop/Query processing/22.py (3.12.10)
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt

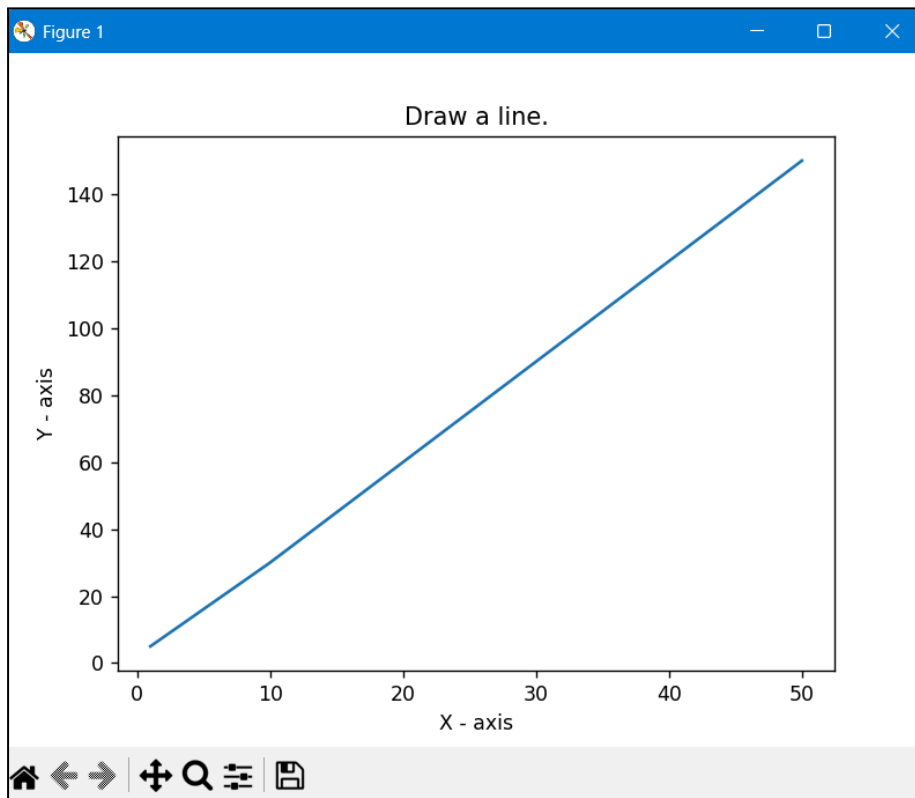
# Data
x = [1, 10, 20, 30, 40, 50]
y = [5, 30, 60, 90, 120, 150]

# Draw the line
plt.plot(x, y)

# Add labels and title
plt.xlabel("X - axis")
plt.ylabel("Y - axis")
plt.title("Draw a line.")

# Display graph
plt.show()
```

OUTPUT



RESULT

Thus, a line graph with suitable x-axis label, y-axis label, and title was successfully created using Matplotlib.

Experiment 23: Draw a Line Using Values from a Text File

Aim

To write a Python program to draw a line using axis values taken from a text file, with suitable labels for the x-axis, y-axis, and a title.

Procedure

- Create a text file containing x-axis and y-axis values.
- Open and read the values from the text file.
- Separate the x and y values.
- Plot the values using Matplotlib.
- Add suitable labels for the x-axis and y-axis.
- Add a title to the graph.
- Display the line graph.

Code

```
23.py - C:/Users/subha/OneDrive/Desktop/Query processing/23.py (3.12.10)
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt

# Read data from text file
x = []
y = []

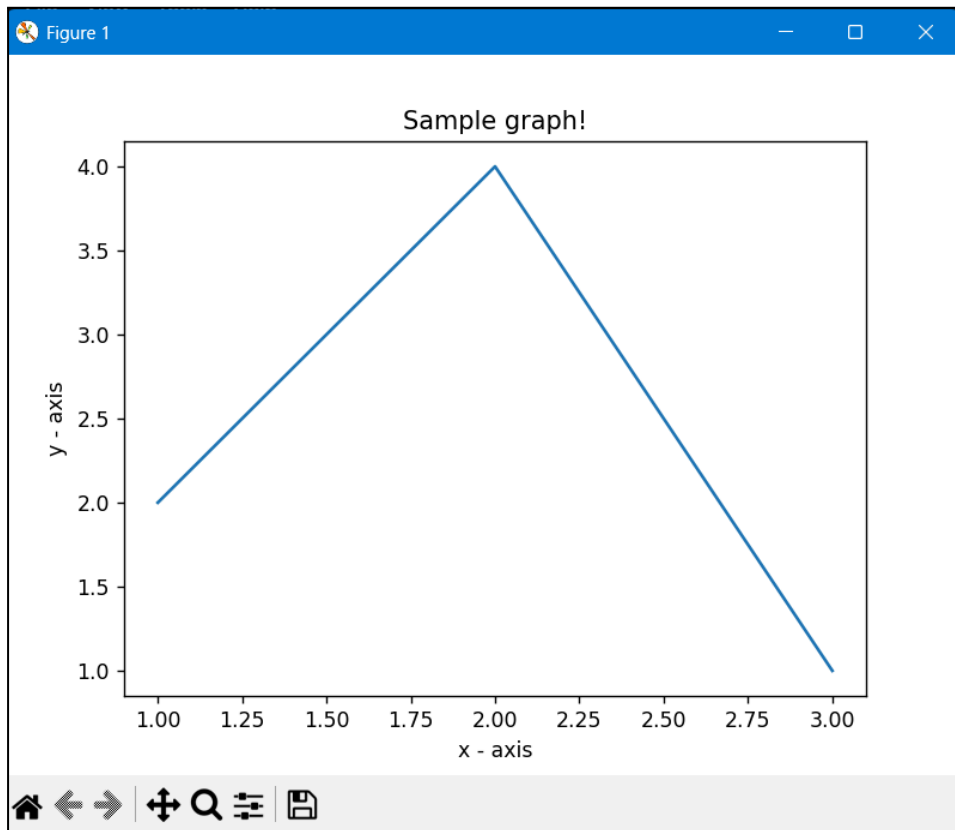
with open("test.txt", "r") as file:
    for line in file:
        values = line.split()
        x.append(float(values[0]))
        y.append(float(values[1]))

# Draw the line
plt.plot(x, y)

# Add labels and title
plt.xlabel("x - axis")
plt.ylabel("y - axis")
plt.title("Sample graph!")

# Display graph
plt.show()
```

Output



Result

Thus, a line graph was successfully drawn using x-axis and y-axis values read from a text file.